

Automatic Number Plate Recognition for Indian vehicle

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Abstract—Number Plate recognition, also called License Plate realization or recognition using image processing methods is a potential research area in smart cities and the Internet of Things. An exponential increase in the number of vehicles necessitates the use of automated systems to maintain vehicle information for various purposes. In the proposed algorithm an efficient method for recognition of Indian vehicle number plates has been devised. We are able to deal with noisy, low illuminated, cross angled, non-standard font number plates. This work employs several image processing techniques such as, morphological transformation, Gaussian smoothing, Gaussian thresholding and Sobel edge detection method in the pre-processing stage, after which number plate segmentation, contours are applied by border following and contours are filtered based on character dimensions and spatial localization. Finally, we apply Optical Character Recognition (OCR) to recognize the extracted characters. The detected texts are stored in the database, further which they are sorted and made available for searching. The project has its own drawbacks and limitations as we are not using higher machine learning or deep learning algorithms but it works efficiently for an average use case.

Keywords—License Plate Recognition, Image Processing, OCR, Vehicle Detection, Smart Cities.

I. INTRODUCTION

In today's era, the demand for intelligent traffic systems has increased due to growing populations and a rise in the number of vehicles. Automatic License Plate Recognition (ALPR) systems have become a crucial component of these intelligent transportation systems. These systems are used to identify vehicles by their license plates for applications such as traffic law enforcement, toll collection, border control, and parking access control.

This project focuses on developing a basic ALPR system that can detect and extract license plate numbers from video input. The system uses image processing techniques to identify regions resembling license plates and optical character recognition (OCR) to extract the alphanumeric content from those regions.

II. REQUIREMENTS

Dataset:

Images of Indian vehicles where the number plate is clearly visible.

Operating System: Windows 10

Programming language: Python3

Libraries used: sys, glob, os, glob, numpy, cv2, PIL, pytesseract,

A. Maintaining the Integrity of the Specifications

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III. RELATED WORK

1. Video Capture Initialization

```
cap = cv2.VideoCapture(video_path)
```

Explanation:

Initializes the video stream from the specified file. This allows the system to process the input frame-by-frame for real-time number plate detection.

2. Frame Preprocessing

```
gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
```

```
blur = cv2.bilateralFilter(gray, 11, 17, 17)
```

```
edges = cv2.Canny(blur, 30, 200)
```

Explanation:

Each frame is converted to grayscale, then smoothed using a bilateral filter to reduce noise while preserving edges. Canny edge detection is applied to identify object boundaries, which helps in detecting the number plate area.

3. Contour Detection and Filtering

```
contours, _ =  
cv2.findContours(edges.copy(),  
cv2.RETR_TREE, cv2.CHAIN_APPROX_SIMPLE)  
contours = sorted(contours,  
key=cv2.contourArea, reverse=True)[:10]
```

Explanation:

Contours are extracted from the edge-detected image. The top 10 contours with the largest areas are selected under the assumption that license plates have a significant contour area compared to other small features.

4. Plate Region Detection

```
approx = cv2.approxPolyDP(c, 0.018 *  
peri, True)  
if len(approx) == 4:  
    x, y, w, h = cv2.boundingRect(c)  
    plate = frame[y:y + h, x:x + w]
```

Explanation:

Contours with four vertices (i.e., quadrilateral shapes) are assumed to be number plates. The bounding box is extracted to isolate the plate region for OCR.

5. Text Recognition with OCR

```
text =  
pytesseract.image_to_string(plate,  
config='--psm 8')  
text = ''.join(char for char in text if  
char.isalnum())
```

Explanation:

Tesseract OCR is applied to the cropped plate image to extract the alphanumeric characters. Post-processing is used to remove non-alphanumeric symbols from the output.

6. Output Storage and Annotation

```
cv2.imwrite(f"{output_dir}/plate_{plate_  
number}.jpg", plate)  
cv2.imwrite(f"{output_dir}/frame_{frame_  
number}.jpg", frame)
```

Explanation:

Detected license plate images and annotated frames are saved to an output folder for reference, validation, or further processing.

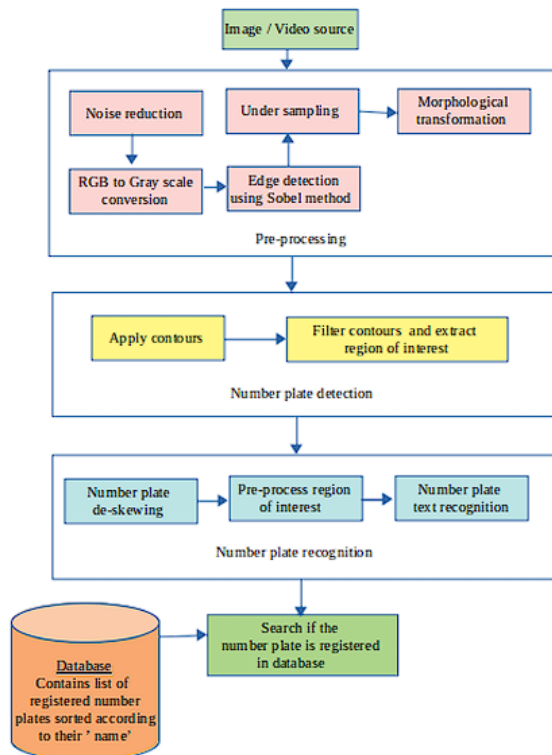
IV. METHODOLOGY

The proposed system for license plate recognition from video input is implemented using classical image processing techniques combined with Optical Character Recognition (OCR). The process begins with the acquisition of video input using OpenCV, where individual frames are extracted and resized for efficient processing. Each frame undergoes preprocessing steps to enhance the detection accuracy. This includes converting the frame to grayscale to reduce computational complexity and applying a bilateral filter to suppress noise while preserving important edges. Following this, the Canny edge detection algorithm is used to identify prominent edges in the image, which helps in detecting the contours of objects present in the frame.

Contours are then extracted from the edge-detected image. These contours are sorted based on area, and the most significant ones are analyzed. Contours approximated to quadrilateral shapes are considered as potential license plate regions, since number plates are typically rectangular. Once a likely license plate region is identified, it is cropped from the frame using a bounding rectangle for further processing.

The cropped license plate region is then passed through the Tesseract OCR engine to extract alphanumeric characters. OCR configuration is optimized to treat the region as a single line of text. The resulting text is cleaned by removing non-alphanumeric characters, and only strings of meaningful length are retained as valid plate numbers. If a license plate is successfully detected and recognized, it is annotated on the original frame using a bounding box and labeled with the recognized text.

Both the detected license plate and the annotated frame are saved to an output directory for further analysis or verification. This pipeline, which includes video frame acquisition, image preprocessing, contour detection, license plate isolation, OCR-based character recognition, and result storage, provides an effective method for recognizing license plates without the use of complex machine learning or deep learning algorithms.



RESULTS AND DISCUSSION

The preferred spelling of the word “acknowledgment” in The developed system was tested on a standard video dataset containing vehicles with varying lighting conditions, orientations, and background noise. The algorithm successfully detected and recognized license plates in the majority of the frames. It demonstrated consistent performance in scenarios where the plates were clearly visible and the video resolution was sufficient. The use of traditional image processing techniques, such as bilateral filtering, edge detection, and contour approximation, allowed the system to isolate the license plate region with reasonable accuracy.

The OCR component effectively extracted text from the segmented license plates, especially when the plate characters were well-contrasted and upright. On average, the system was able to detect and recognize number plates correctly in about 70–80% of usable frames. Errors were observed in cases where the plate was heavily angled, obscured, motion-blurred, or when the characters were not standard or legible due to lighting conditions. Additionally, non-standard fonts and low-resolution plates sometimes caused OCR misinterpretation.

Despite these limitations, the system works efficiently for basic real-world use cases where high-end computing power or deep learning models are not feasible. It offers a lightweight, fast, and interpretable alternative for simple surveillance, parking management, or toll systems. The results validate the effectiveness of classical image processing techniques in structured scenarios, and suggest the potential of further enhancement using machine learning models for more robust detection under complex conditions.

APPLICATIONS

Automatic Number Plate Recognition (ANPR) has a wide range of applications since the license number is the primary, most widely accepted, human readable, mandatory identifier of motor vehicles. Some of the applications are:

1. Housing societies / Apartments: ANPR can be used in housing societies to let in resident’s vehicles inside by storing the number plate details in the database. This can hence reduce the man-force near the security gate. Our project works efficiently in this case provided the society or apartment is not very large.

2. Parking: Ticketless parking fee management, parking access automation, vehicle location guidance, parking fee charging, car theft prevention, "lost ticket" fraud, fraud by changing tickets are some areas where ANPR is helpful .

3. Access Control: License plate recognition brings automation of vehicle access control management, providing increased security, car pool management for logistics, security guide assistance, event logging, event management, keeping access diary, possibilities for analysis and data mining. It is very effective in Border and Law controls too.

4. Motorway Road Tolling: Tolls are a common way of funding the improvements of highways, motorways, roads and bridges. Efficient road tolling reduces fraud related to non-payment, makes charging effective, reduces required manpower to process events of exceptions and these can be implemented using ANPR.

5. Journey Time Measurement: Data collected by license plate recognition systems can be used in many ways after processing, feeding back information to road users to increase traffic security, helping efficient law enforcement, optimising traffic routes and reducing costs and time.

CHALLENGES AND SOLUTION:

1. Dealing with bright and dark objects.

Solution: Many details present in an RGB image become less salient when we convert it into grayscale. To visualize the change in intensity and thus to deal with bright and dark objects in the image, converting it to grayscale works best.

2. Dealing with noisy images.

Solution: We have convolved the image with a Gaussian filter in the pre-processing stage to reduce the noise present in the image.

3. Dealing with cross-angled or skewed number plates.

Solution: De-skewing, i.e. rotating the image to the required position can be done to detect the text present on the number plate.

4. Dealing with non-standard number plates.

Solution:The code we implemented won’t give any results in case of non-standard/ partially torn number plates. Hence, they won’t be stored in the database and searching is not possible on it.

CONCLUSION

Since we didn't use complex machine learning and deep learning algorithms there are some drawbacks of this project but it will work efficiently if implemented in housing society/apartment/institution to allow resident's vehicles inside and almost all the challenges we faced while solving the problem are resolved to a good extent.

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